

Practice test - Programming for Minor Computational Science

Name:

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Log in:

- You will take this exam using the online editor. To access it:
 - Go to **mcs.proglab.nl/exams**

Goal:

- This digital exam consists of 3 assignments where you will write short Python programs.
- You will be **evaluated solely on the correctness** of your solutions. Code design, comments, and style are not important, so you do not need to worry about them.
- The editor has a Checkpy button that allows you to verify your solutions during the exam for your personal use. Note that Checkpy results will not influence your actual grade, as all grading is done manually. So, even if your solutions do not pass Checkpy, they may still be accepted during the grading process.

Constraints:

- You may use the course website as a resource (open mcs.proglab.nl again in a new tab after login in). Keep in mind that:
 - you cannot use any other website,
 - you cannot rely on any existing code you've written before this exam,
 - you cannot get assistance with programming during the exam.
- You may use a printed version of the cheatsheet during the exam.
- Do not use external modules such as `numpy`, `csv`, or others unless the assignment specifically says you can.

Submit:

- When you have completed all assignments:
 - use the **submit button** on the website to ensure your final edits are committed,
 - close the browser.

Assignment 1: Odd

Write a function named `sumodd(numbers)` that accepts a list of integers and returns the sum of all odd numbers in the list.

Example Usage:

```
numbers = [1, 2, 3, 1, 2, 3, 3, 4]
result = sumodd(numbers)
print(result)
```

Expected Output:

```
11
```

Tip: you can use `number % 2 == 1` to test if a number is odd.

Assignment 2: Reverse

Write a function named `reverse_list(lst)` that accepts a list and returns a new list with the elements in reverse order.

Example Usage:

```
lst = ["he", "l", "lo"]
result = reverse_list(lst)
print(result)
```

Expected Output:

```
['lo', 'l', 'he']
```

Assignment 3: Reps

Write a function named `longest_repetition(lst)` that accepts a list and returns the length of the longest consecutive repetition of the same element in the list.

For example, in the list `[1, 2, 2, 5, 5, 5, "a", "a"]`, the element 5 appears three times in a row, which is the longest consecutive repetition.

Example Usage:

```
lst = [1, 2, 2, 5, 5, 5, "a", "a"]
result = longest_repetition(lst)
print(result)
```

Expected Output:

```
3
```

Assignment 4: Grid

Write a function named `generate_1x1_grid(step)` that generates all coordinate points within the 1x1 grid (from (0,0) to (1,1)) using the given `step` value. The function should return a list of tuples representing the grid points.

Example Usage:

```
result = generate_1x1_grid(0.5)
print(result)
```

Expected Output:

```
[(0.0, 0.0), (0.0, 0.5), (0.0, 1.0), (0.5, 0.0), (0.5, 0.5), (0.5, 1.0), (1.0, 0.0),
(1.0, 0.5), (1.0, 1.0)]
```